Week 3 Session A

Tuesday, Jan 26 2026

Overview

- Review: The two phases of a LLM & two pipelines of a LLM
- Setting Up and Exploring Models Together:
  "Where does AI actually run?"
  - The Physicality of AI Systems
- Week 3 Assignment

Review

- LLMs have two phases:
  - Phase A
  - Phase B

- Ture or False?
  - Tokenization happens at both phases!
  - Tokens and characters are pretty much the same thing.

Review

- LLMs don’t retrieve information, they generate!
- BUT gen AI tools are different! They can do both! RAG (Retrieval-Augmented Generation) helps optimize the output of LLMs.

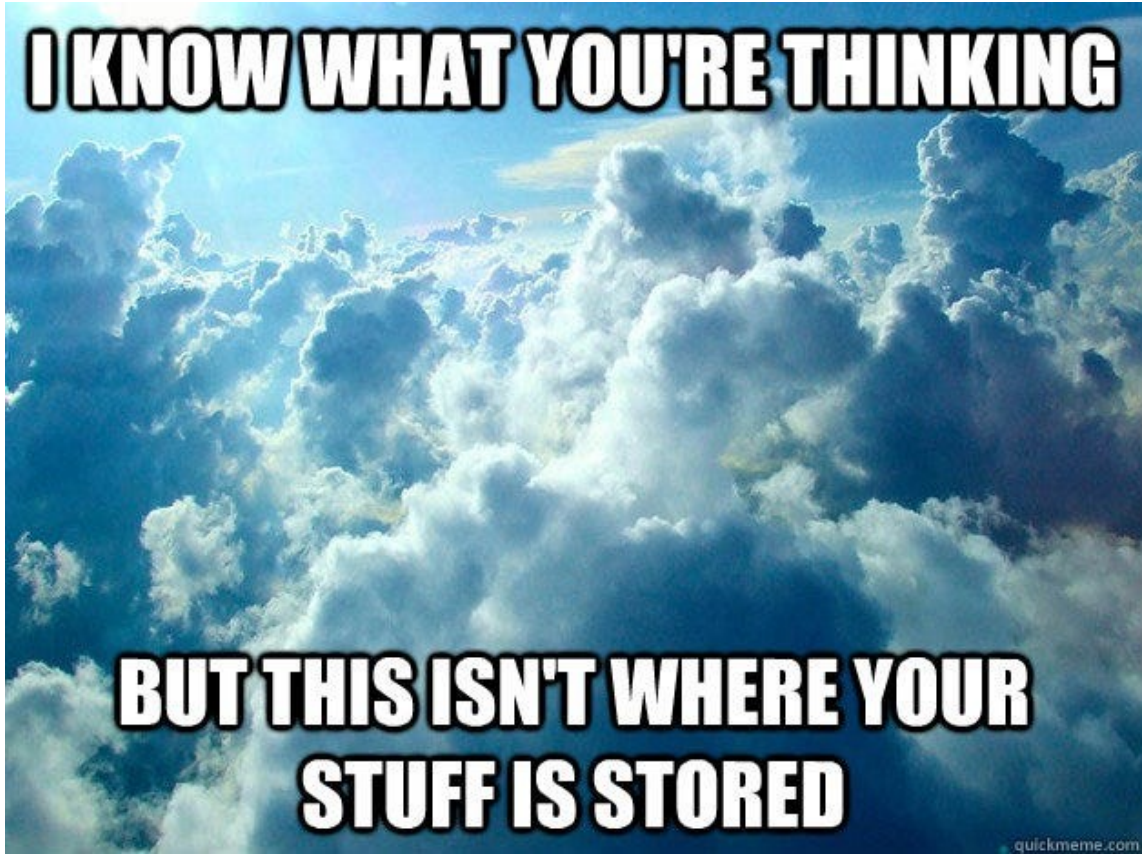
“Where does AI actually run?”

- Activity I:
  - Draw a visual representation of this statement “most people use AI through the cloud.”
  - Draw what you think happens when you ask a genAI tool (like ChatGPT, Copilot, etc.) a question.

What does the “Cloud” actually mean?

- Activity II:
  - Search the word “Cloud” in your preferred browser. What are some key terms you are seeing?
  - Work with each other, and come up with a definition for cloud in the context of this class.

What does the “Cloud” actually mean?



What does the “Cloud” actually mean?

- From my point of view:
  - Cloud is <u>someone else</u>’s <u>computer/server</u> (server is a powerful computer), and I can have access to via internet.
  - There are multiple Cloud providers (Google, Microsoft, etc.).
  - Some terms: data center, server, cloud computing, etc.
  - Bottom line: GPT-5.2 doesn’t live in the air; it lives in a box (or racks of boxes) of GPUs somewhere in Iowa (maybe).

“Where does AI actually run?”

- So what happens when I actually send a question to a genAI chatbot like ChatGPT?

Let’s Map Out the Ecosystem

- Activity:
  Build a physical map of the journey your prompt takes: from your laptop all the way to the machine that runs the model. Place these cards on the table in that order.

Let’s Map Out the Ecosystem

- From my point of view:
  AI does have carbon footprint.
  People build it; people maintain it. It costs $, water, labor.
  Nothing lives “in the cloud”. It lives in a building with machines.

Oh wait… what does MSU have?

- ChatGPT runs far away in Des Moines Iowa. But MLStudio that we are going to use runs right here at MSU on ICER's hardware.
- MSU Campus HPC vs Commercial Cloud:
  - Shorter network path
  - Smaller and open models
  - Job scheduled for specific GPUs
  - Local policies for power and cooling

LMStudio!

- ICER MLStudio on Campus HPCC

MSU HPCC OnDemand | Files | Jobs | Interactive Apps | Development Nodes

Home / My Interactive Sessions / LM Studio (BETA)

Interactive Apps

Desktops

Interactive Desktop

GUIs

LM Studio (BETA)

MATLAB

OpenWebUI (BETA)

ParaView

Stata

Storage Analyzer

Servers

Code Server

Jupyter

RStudio Server

LM Studio (BETA)

Discover, download, and run local LLMs.

NOTE: This app is currently in beta testing and may not work as intended. Please contact us with any questions or issues.

☐ I would like to receive an email when the session starts

Number of hours

1

Jobs shorter than four hours will schedule much faster

Number of cores

4

If you use multiple tasks (see Advanced Options), this is the number of cores per task

Amount of memory

4

Input a number of GB with **no suffix**, e.g., `100` for 100GB or `1000` for 1TB. Minimum of `2` GB per core recommended.

☑ Use GPUs (recommended)

Week 3 Assignment

- Select a language model on LMStudio, explain why you picked this one (short rationale + screenshot).